FORM A

PROJECT IDEA SUBMISSION FORM
INTEGRATED ENGINEERING TEAM PROJECT (IETP4115)

Project Title: AquaPave: Permeable Pavement with Built-in Filtration and a Clogging Alert

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Collaborator(s) (if any):

Project concept and SDG mapping:

Problem: City roads and sidewalks are sealed with concrete or asphalt. Rain cannot soak in, so it floods streets, carries dirt and oil into drains, and the water is wasted.

Concept: A pavement that lets rain soak through pervious concrete, then through a sand and charcoal (or pumice) filter layer and a gravel storage layer. A perforated pipe carries the cleaned water to a tank for reuse. Water-level, rain and turbidity sensors with an ESP32 detect when drainage slows because of clogging and send a cleaning alert.

SDG mapping:

- SDG 11 Sustainable Cities and Communities (primary): reduces street flooding and makes city streets more resilient.
- SDG 6 Clean Water and Sanitation: filters runoff and allows the water to be reused.
- SDG 9 Industry, Innovation and Infrastructure: low-cost, resilient infrastructure with smart monitoring.

Departments involved: Civil, Chemical, Environmental, Architecture, Electromechanical, Electrical and Computer, and Software engineering.

Objectives:

1. To design and build a prototype permeable pavement with pervious concrete, a filter layer, a gravel storage layer and a perforated drainage pipe.
2. To test the infiltration rate and the percentage of rain that runs off, compared with a normal concrete slab.
3. To measure how well the filter layer cleans the water, using turbidity before and after filtering.
4. To develop a monitoring system (water level, rain and turbidity) that detects clogging and sends a cleaning alert.
5. To design a sidewalk or plaza with permeable zones and build a scale model.

Short summary of the project (not more than 200 words)

City roads and sidewalks are sealed with concrete or asphalt, so rain cannot soak into the ground. This causes street flooding, carries dirt and oil into drains, and wastes water. AquaPave is a permeable pavement that lets rain soak through and cleans it on the way down. The top layer is pervious concrete made of cement and single-size gravel with no sand. Below it, a sand layer with charcoal or pumice traps dirt and some pollutants, and a gravel layer stores the water temporarily. A perforated pipe carries the cleaned water to a tank for reuse. An ESP32 reads a water-level sensor, a rain sensor and a turbidity sensor. If drainage becomes slower over time, the code detects clogging and sends a cleaning alert. The prototype will be tested on sloped trays, using a watering can as rain, and compared with a normal concrete slab. We will measure infiltration rate, runoff percentage and turbidity before and after filtering, and check that the alert triggers correctly. A scale model of a sidewalk with permeable zones shows how it could be used in a city. The project brings together seven engineering departments.

Materials, Tools, equipment/instruments required:

- Pavement: cement, single-size gravel, water, molds, sand, charcoal or pumice, gravel, geotextile or mesh.
- Test rig: two sloped trays or containers, a collection tank, perforated PVC pipe and fittings, a watering can or small sprinkler.
- Electronics: ESP32, water-level sensor (ultrasonic or float), rain sensor, turbidity sensor, LCD, wiring.
- Power: USB or 5 V supply (optional small solar panel).
- Scale model: foam board, cardboard or plywood.
- Tools and instruments: measuring cylinder, stopwatch, weighing scale, mixing tools, drilling and cutting tools, multimeter.